CS6650 Assignment 1

GitHub Repository: https://github.com/Zixin-Y/cs6650-assignment1

Design and Performance Evaluation

Architecture

The Java 11 load client runs on a local Mac and communicates with a Tomcat 9 server deployed on EC2.

HTTP flow:

SwipeGenerator → BlockingQueue → sender threads →shared HttpClient → EC2 SkierServlet.

WebSocket flow:

EC2 LeaderboardEndpoint → ten persistent LeaderboardSubscriber instances → client metrics.

SkierServlet validates request paths and JSON bodies. Valid ski events receive HTTP 201; invalid requests receive HTTP 400. HealthServlet provides a health-check endpoint.

LeaderboardEndpoint publishes synthetic leaderboard messages approximately every 500 ms. These messages are independent of the submitted events. No database or event aggregation is used in this assignment.

Classes and Threading

SwipeEvent represents one generated ski event.

SwipeGenerator runs on one dedicated thread and places 500,000 events into an ArrayBlockingQueue. Generation finishes before sender execution, so senders do not wait for the generator during the measured ingestion phases.

IngestLoadClient coordinates generation, subscriptions, warmup, the main phase, shutdown, and reporting.

Warmup uses 32 sender threads to send 32,000 events. The main phase sends the remaining 468,000 events. Main thread counts of 64, 128, and 256 were tested.

Each sender task receives a fixed event quota, polls events from the shared queue, and calls SwipeSender. The coordinator waits for all tasks to finish before ending a phase.

SwipeSender shares one HTTP/1.1 HttpClient across sender threads, allowing persistent connection reuse. It uses thread-safe counters and retries unsuccessful requests up to five times with exponential backoff starting at 100 ms.

WebSocket Strategy

Ten LeaderboardSubscriber instances subscribe to resorts 1 through 10 using a shared subscription HttpClient.

Each subscriber has a background worker that establishes the connection, waits for disconnection, and reconnects with backoff. Message callbacks assemble fragmented text, parse JSON, and record update counts and timestamp differences.

Ingestion starts after every subscriber has received an update. Subscribers remain active throughout ingestion and are closed during cleanup.

Metrics include connection attempts, successful connections, reconnections, connection establishment time, and updates.

Measurement Design

SwipeCsvRecorder records each HTTP attempt with timestamp, request type, latency, status code, and resort ID. Writes are synchronized. Network failures use status code zero.

HTTP durations use System.nanoTime(), avoiding dependence on clock alignment between the client and server.

LatencyStats computes mean, median, nearest-rank p95 and p99, minimum, and maximum. SwipeCsvAnalysis reports attempt counts, latency statistics, and successful throughput by resort.

ThroughputBuckets groups successful request completions into 10-second intervals. The final partial interval uses its actual duration. Completion times are approximated from the attempt timestamp plus measured latency.

Generation, warmup, main ingestion, and total runtime are reported separately. Main throughput excludes warmup.

Little's Law and Prediction

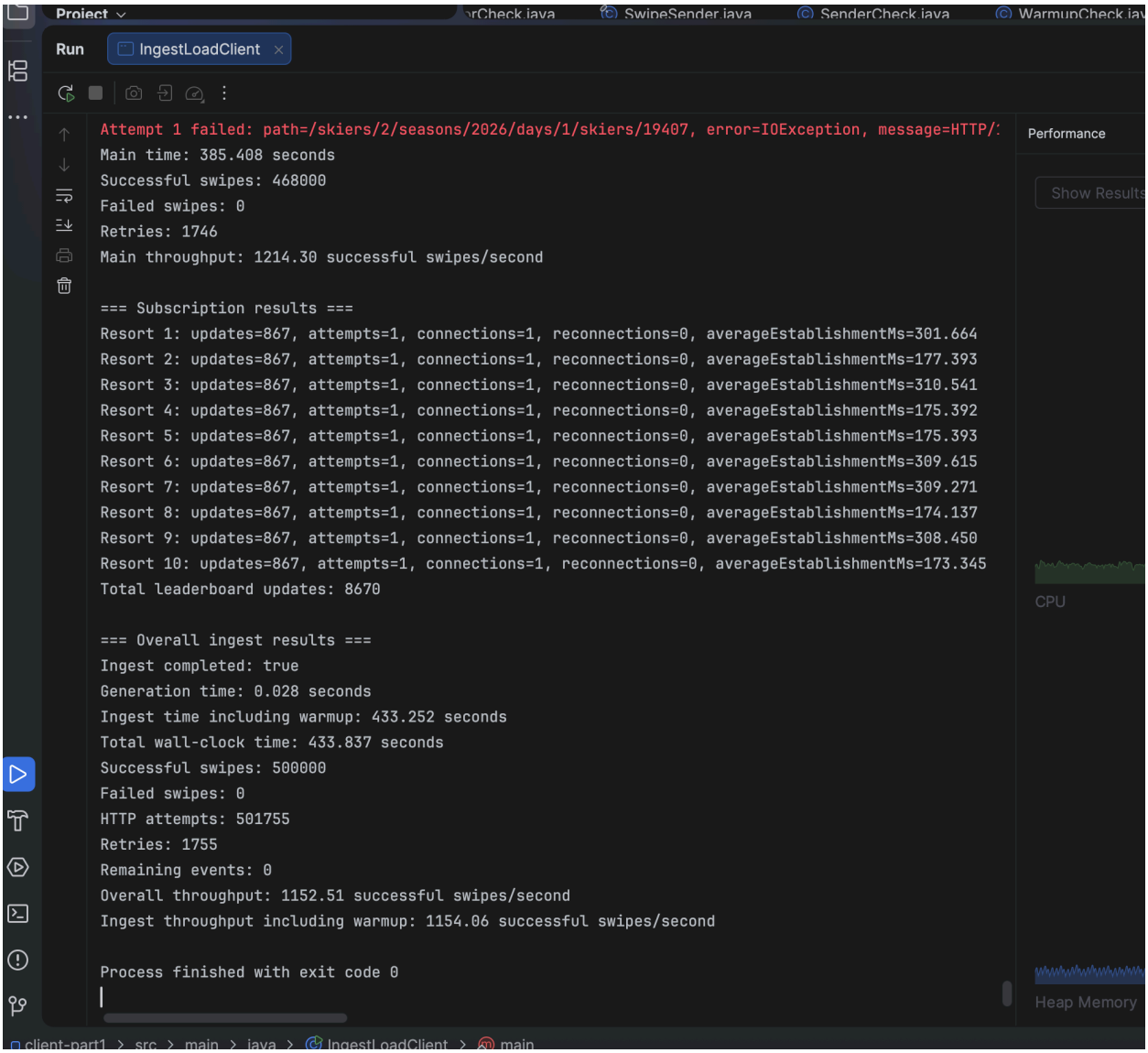
The measured sequential HTTP baseline mean was 37.171 ms.

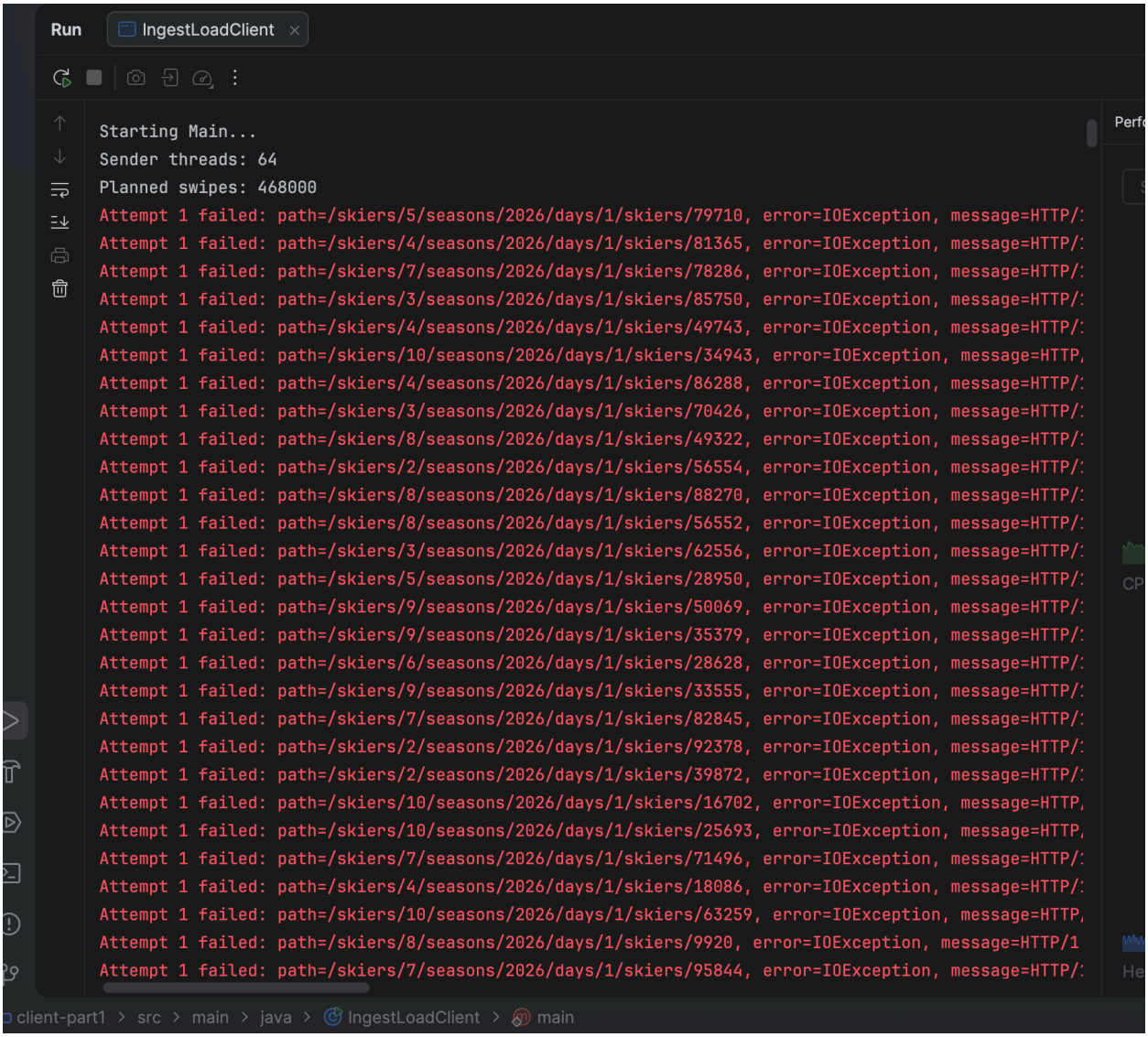
Using L = lambda × W, with L approximated by the number of synchronous sender threads:

Predicted throughput = sender threads / 0.037171 seconds.

Predictions are approximately: 32 threads: 861 events/s; 64 threads: 1,722 events/s; 128 threads: 3,444 events/s; 256 threads: 6,887 events/s.

Part 1 Results

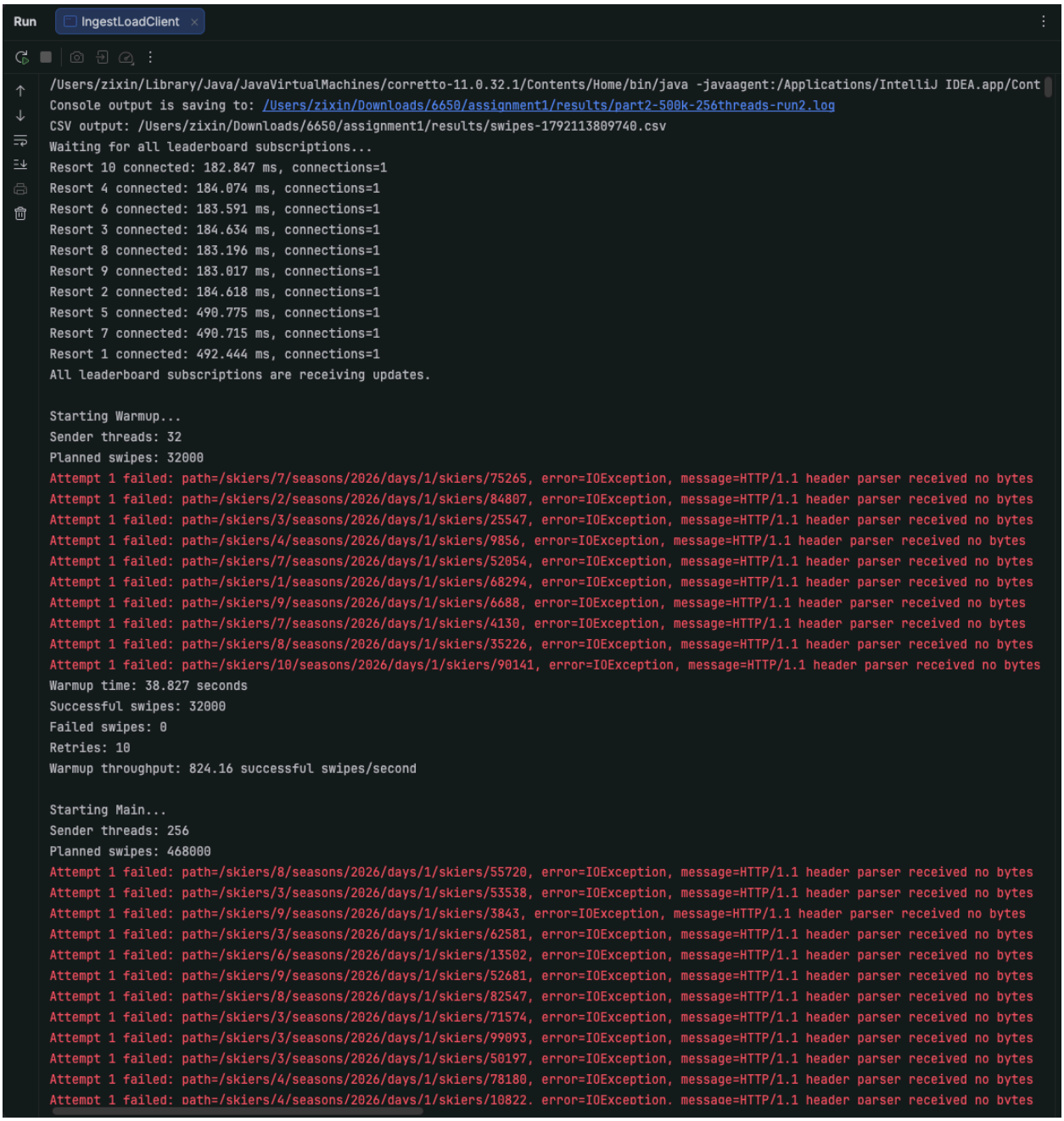



Run
IngestLoadClient
Starting Main...
Sender threads: 64
Planned swipes: 468000
Attempt 1 failed: path=/skiers/5/seasons/2026/days/1/skiers/79710, error=IOException, message=HTTP/1
Attempt 1 failed: path=/skiers/4/seasons/2026/days/1/skiers/81365, error=IOException, message=HTTP/1
Attempt 1 failed: path=/skiers/7/seasons/2026/days/1/skiers/78286, error=IOException, message=HTTP/1
Attempt 1 failed: path=/skiers/3/seasons/2026/days/1/skiers/85750, error=IOException, message=HTTP/1
Attempt 1 failed: path=/skiers/4/seasons/2026/days/1/skiers/49743, error=IOException, message=HTTP/1
Attempt 1 failed: path=/skiers/10/seasons/2026/days/1/skiers/34943, error=IOException, message=HTTP,
Attempt 1 failed: path=/skiers/4/seasons/2026/days/1/skiers/86288, error=IOException, message=HTTP/1
Attempt 1 failed: path=/skiers/3/seasons/2026/days/1/skiers/70426, error=IOException, message=HTTP/1
Attempt 1 failed: path=/skiers/8/seasons/2026/days/1/skiers/49322, error=IOException, message=HTTP/1
Attempt 1 failed: path=/skiers/2/seasons/2026/days/1/skiers/56554, error=IOException, message=HTTP/1
Attempt 1 failed: path=/skiers/8/seasons/2026/days/1/skiers/88270, error=IOException, message=HTTP/1
Attempt 1 failed: path=/skiers/8/seasons/2026/days/1/skiers/56552, error=IOException, message=HTTP/1
Attempt 1 failed: path=/skiers/3/seasons/2026/days/1/skiers/62556, error=IOException, message=HTTP/1
Attempt 1 failed: path=/skiers/5/seasons/2026/days/1/skiers/28950, error=IOException, message=HTTP/1
Attempt 1 failed: path=/skiers/9/seasons/2026/days/1/skiers/50069, error=IOException, message=HTTP/1
Attempt 1 failed: path=/skiers/9/seasons/2026/days/1/skiers/35379, error=IOException, message=HTTP/1
Attempt 1 failed: path=/skiers/6/seasons/2026/days/1/skiers/28628, error=IOException, message=HTTP/1
Attempt 1 failed: path=/skiers/9/seasons/2026/days/1/skiers/33555, error=IOException, message=HTTP/1
Attempt 1 failed: path=/skiers/7/seasons/2026/days/1/skiers/82845, error=IOException, message=HTTP/1
Attempt 1 failed: path=/skiers/2/seasons/2026/days/1/skiers/92378, error=IOException, message=HTTP/1
Attempt 1 failed: path=/skiers/2/seasons/2026/days/1/skiers/39872, error=IOException, message=HTTP/1
Attempt 1 failed: path=/skiers/10/seasons/2026/days/1/skiers/16702, error=IOException, message=HTTP,
Attempt 1 failed: path=/skiers/10/seasons/2026/days/1/skiers/25693, error=IOException, message=HTTP,
Attempt 1 failed: path=/skiers/7/seasons/2026/days/1/skiers/71496, error=IOException, message=HTTP/1
Attempt 1 failed: path=/skiers/4/seasons/2026/days/1/skiers/18086, error=IOException, message=HTTP/1
Attempt 1 failed: path=/skiers/10/seasons/2026/days/1/skiers/63259, error=IOException, message=HTTP,
Attempt 1 failed: path=/skiers/8/seasons/2026/days/1/skiers/9920, error=IOException, message=HTTP/1
Attempt 1 failed: path=/skiers/7/seasons/2026/days/1/skiers/95844, error=IOException, message=HTTP/1
client-part1 > src > main > java > IngestLoadClient > main

```
/Users/zixin/Library/Java/JavaVirtualMachines/corretto-11.0.32.1/Contents/Home/bin/java -javaagent:,
Waiting for all leaderboard subscriptions...
Resort 4 connected: 175.392 ms, connections=1
Resort 5 connected: 175.393 ms, connections=1
Resort 2 connected: 177.393 ms, connections=1
Resort 8 connected: 174.137 ms, connections=1
Resort 10 connected: 173.345 ms, connections=1
Resort 1 connected: 301.664 ms, connections=1
Resort 3 connected: 310.541 ms, connections=1
Resort 7 connected: 309.271 ms, connections=1
Resort 6 connected: 309.615 ms, connections=1
Resort 9 connected: 308.450 ms, connections=1
All leaderboard subscriptions are receiving updates.

Starting Warmup...
Sender threads: 32
Planned swipes: 32000
Attempt 1 failed: path=/skiers/10/seasons/2026/days/1/skiers/25783, error=IOException, message=HTTP,
Attempt 1 failed: path=/skiers/2/seasons/2026/days/1/skiers/76358, error=IOException, message=HTTP/1
Attempt 1 failed: path=/skiers/9/seasons/2026/days/1/skiers/41595, error=IOException, message=HTTP/1
Attempt 1 failed: path=/skiers/8/seasons/2026/days/1/skiers/83565, error=IOException, message=HTTP/1
Attempt 1 failed: path=/skiers/7/seasons/2026/days/1/skiers/64947, error=IOException, message=HTTP/1
Attempt 1 failed: path=/skiers/5/seasons/2026/days/1/skiers/58033, error=IOException, message=HTTP/1
Attempt 1 failed: path=/skiers/6/seasons/2026/days/1/skiers/4885, error=IOException, message=HTTP/1
Attempt 1 failed: path=/skiers/3/seasons/2026/days/1/skiers/84785, error=IOException, message=HTTP/1
Attempt 1 failed: path=/skiers/10/seasons/2026/days/1/skiers/83275, error=IOException, message=HTTP,
Warmup time: 47.838 seconds
Successful swipes: 32000
Failed swipes: 0
Retries: 9
Warmup throughput: 668.93 successful swipes/second
```

Part 2 Results — 256 Threads, Run 2




Run
IngestLoadClient
Performance
Attempt 1 failed: path=/skiers/7/seasons/2026/days/1/skiers/95125, error=HttpTimeoutException, message=request timed out
Attempt 1 failed: path=/skiers/5/seasons/2026/days/1/skiers/994, error=IOException, message=HTTP/1.1 header parser received no bytes
Attempt 1 failed: path=/skiers/8/seasons/2026/days/1/skiers/3206, error=IOException, message=HTTP/1.1 header parser received no bytes
Attempt 1 failed: path=/skiers/6/seasons/2026/days/1/skiers/14896, error=IOException, message=HTTP/1.1 header parser received no bytes
Main time: 100.531 seconds
Successful swipes: 468000
Failed swipes: 0
Retries: 1876
Main throughput: 4655.29 successful swipes/second
=== Subscription results ===
Resort 1: updates=279, attempts=1, connections=1, reconnections=0, averageEstablishmentMs=492.444
Resort 2: updates=279, attempts=1, connections=1, reconnections=0, averageEstablishmentMs=184.618
Resort 3: updates=279, attempts=1, connections=1, reconnections=0, averageEstablishmentMs=184.634
Resort 4: updates=279, attempts=1, connections=1, reconnections=0, averageEstablishmentMs=184.074
Resort 5: updates=279, attempts=1, connections=1, reconnections=0, averageEstablishmentMs=490.775
Resort 6: updates=279, attempts=1, connections=1, reconnections=0, averageEstablishmentMs=183.591
Resort 7: updates=279, attempts=1, connections=1, reconnections=0, averageEstablishmentMs=490.715
Resort 8: updates=279, attempts=1, connections=1, reconnections=0, averageEstablishmentMs=183.196
Resort 9: updates=279, attempts=1, connections=1, reconnections=0, averageEstablishmentMs=183.017
Resort 10: updates=279, attempts=1, connections=1, reconnections=0, averageEstablishmentMs=182.847
Total leaderboard updates: 2790
Resort 1 push latency: count=279, meanMs=20.884, medianMs=20.440, p95Ms=26.641, p99Ms=31.959, minMs=15.213, maxMs=42.970
Resort 2 push latency: count=279, meanMs=21.062, medianMs=20.523, p95Ms=26.236, p99Ms=32.009, minMs=15.722, maxMs=43.179
Resort 3 push latency: count=279, meanMs=21.227, medianMs=20.595, p95Ms=26.738, p99Ms=35.645, minMs=16.986, maxMs=50.834
Resort 4 push latency: count=279, meanMs=21.358, medianMs=20.677, p95Ms=26.858, p99Ms=35.831, minMs=16.823, maxMs=51.045
Resort 5 push latency: count=279, meanMs=18.303, medianMs=17.624, p95Ms=25.137, p99Ms=31.679, minMs=13.148, maxMs=42.070
Resort 6 push latency: count=279, meanMs=18.194, medianMs=17.443, p95Ms=24.743, p99Ms=31.511, minMs=13.529, maxMs=41.860
Resort 7 push latency: count=279, meanMs=18.023, medianMs=17.328, p95Ms=24.564, p99Ms=30.556, minMs=13.477, maxMs=33.409
Resort 8 push latency: count=279, meanMs=21.173, medianMs=20.508, p95Ms=26.620, p99Ms=35.680, minMs=16.661, maxMs=50.323
Resort 9 push latency: count=279, meanMs=21.400, medianMs=20.682, p95Ms=27.557, p99Ms=36.012, minMs=16.893, maxMs=51.072
Resort 10 push latency: count=279, meanMs=18.523, medianMs=17.712, p95Ms=25.110, p99Ms=31.644, minMs=13.669, maxMs=41.615
All subscribers push latency: count=2790, meanMs=20.015, medianMs=19.711, p95Ms=25.966, p99Ms=31.679, minMs=13.148, maxMs=51.072
Negative push-latency samples: 0
Push latency CSV: /Users/zixin/Downloads/6650/assignment1/results/push-latencies-1792113950073.csv
=== Overall ingest results ===
Ingest completed: true
Generation time: 0.032 seconds
Ingest time including warmup: 139.381 seconds
Total wall-clock time: 140.323 seconds
Successful swipes: 500000
Failed swipes: 0
HTTP attempts: 501886
Retries: 1886
Remaining events: 0
Overall throughput: 3563.21 successful swipes/second
Ingest throughput including warmup: 3587.29 successful swipes/second
CSV rows: 501886
CSV file closed successfully.

```
=== HTTP latency and resort analysis ===
Input CSV: /Users/zixin/Downloads/6650/assignment1/results/swipes-1792113809740.csv
HTTP attempts: 501886
Successful responses: 500000
Network-error attempts: 1886
Other HTTP responses: 0

All HTTP attempt latencies: count=501886, meanMs=44.854, medianMs=40.996, p95Ms=65.720, p99Ms=102.729, minMs=1.154, maxMs=10009.410
Successful HTTP attempt latencies: count=500000, meanMs=44.280, medianMs=41.007, p95Ms=65.556, p99Ms=98.263, minMs=17.041, maxMs=630.71

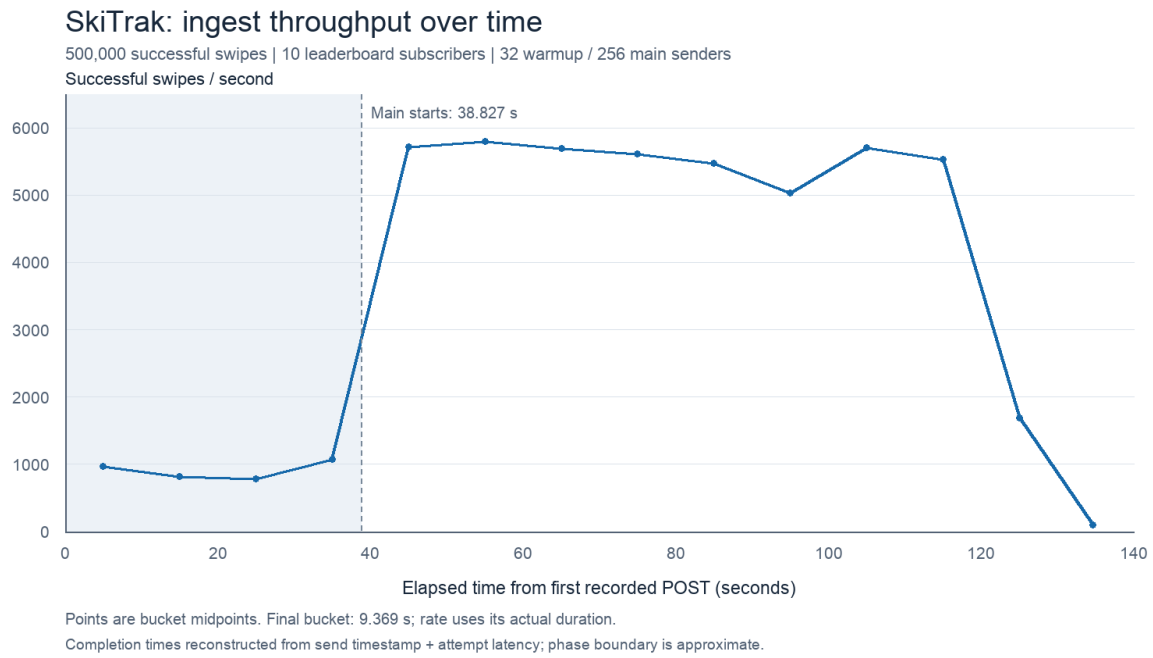
Successful swipes by resort:
Measurement duration: 139.381 seconds
Resort 1: successful=49925, throughput=358.191 swipes/s
Resort 2: successful=49894, throughput=357.969 swipes/s
Resort 3: successful=50054, throughput=359.117 swipes/s
Resort 4: successful=50149, throughput=359.798 swipes/s
Resort 5: successful=49911, throughput=358.091 swipes/s
Resort 6: successful=50260, throughput=360.595 swipes/s
Resort 7: successful=49742, throughput=356.878 swipes/s
Resort 8: successful=49776, throughput=357.122 swipes/s
Resort 9: successful=50162, throughput=359.892 swipes/s
Resort 10: successful=50127, throughput=359.640 swipes/s
Total ingest throughput: 3587.293 swipes/s

=== Throughput buckets ===
0.000,10.000,10.000,9745,974.500
10.000,20.000,10.000,8151,815.100
20.000,30.000,10.000,7881,788.100
30.000,40.000,10.000,10743,1074.300
40.000,50.000,10.000,57173,5717.300
50.000,60.000,10.000,57932,5793.200
60.000,70.000,10.000,56984,5698.400
70.000,80.000,10.000,56108,5610.800
80.000,90.000,10.000,54745,5474.500
90.000,100.000,10.000,50354,5035.400
100.000,110.000,10.000,57011,5701.100
110.000,120.000,10.000,55363,5536.300
120.000,130.000,10.000,16927,1692.700
130.000,139.369,9.369,883,94.249
Successful swipes counted: 500000
Throughput CSV saved to: /Users/zixin/Downloads/6650/assignment1/results/swipes-1792113809740-throughput.csv

Process finished with exit code 0
```

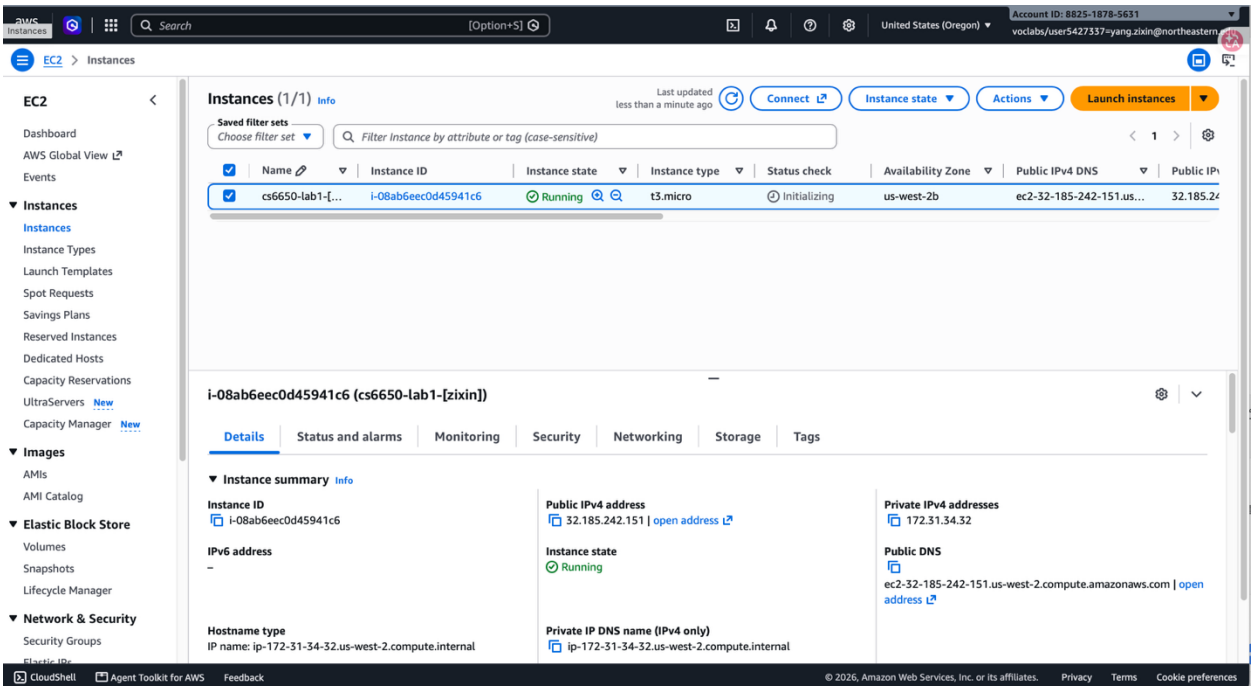
Throughput and Comparison

The second 256-thread run achieved approximately 67.6% of the baseline prediction. The estimate assumes constant response time. Under load, response time, retries, scheduling, logging, and resource limits may change. These measurements do not isolate a single bottleneck.



Main threads	Main throughput (events/s)
64	1495.52
128	2691.50
256-Run1	4632.00
256-Run2	4655.29

Deployment and WebSocket Verification




```
zixin@Zixins-MacBook-Air ski-servlet % npx --yes --package=wscat wscat -c ws://32.185.242.151:8080/ski-servlet/leaderboard/7
Connected (press CTRL+C to quit)
< {"resortID":"7","serverTimestamp":"2026-10-16T21:50:29.060180Z","top":[{"rank":1,"skierID":40213,"vertical":44120},{"rank":2,"skierID":11872,"vertical":43880}]}
< {"resortID":"7","serverTimestamp":"2026-10-16T21:50:29.552319Z","top":[{"rank":1,"skierID":40213,"vertical":44120},{"rank":2,"skierID":11872,"vertical":43880}]}
< {"resortID":"7","serverTimestamp":"2026-10-16T21:50:30.052123Z","top":[{"rank":1,"skierID":40213,"vertical":44120},{"rank":2,"skierID":11872,"vertical":43880}]}
< {"resortID":"7","serverTimestamp":"2026-10-16T21:50:30.552110Z","top":[{"rank":1,"skierID":40213,"vertical":44120},{"rank":2,"skierID":11872,"vertical":43880}]}
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< {"resortID":"7","serverTimestamp":"2026-10-16T21:50:35.052113Z","top":[{"rank":1,"skierID":40213,"vertical":44120},{"rank":2,"skierID":11872,"vertical":43880}]}
< {"resortID":"7","serverTimestamp":"2026-10-16T21:50:35.552124Z","top":[{"rank":1,"skierID":40213,"vertical":44120},{"rank":2,"skierID":11872,"vertical":43880}]}
< {"resortID":"7","serverTimestamp":"2026-10-16T21:50:36.052167Z","top":[{"rank":1,"skierID":40213,"vertical":44120},{"rank":2,"skierID":11872,"vertical":43880}]}
< {"resortID":"7","serverTimestamp":"2026-10-16T21:50:36.552127Z","top":[{"rank":1,"skierID":40213,"vertical":44120},{"rank":2,"skierID":11872,"vertical":43880}]}
< {"resortID":"7","serverTimestamp":"2026-10-16T21:50:37.052125Z","top":[{"rank":1,"skierID":40213,"vertical":44120},{"rank":2,"skierID":11872,"vertical":43880}]}
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< {"resortID":"7","serverTimestamp":"2026-10-16T21:50:38.552124Z","top":[{"rank":1,"skierID":40213,"vertical":44120},{"rank":2,"skierID":11872,"vertical":43880}]}
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< {"resortID":"7","serverTimestamp":"2026-10-16T21:50:39.552133Z","top":[{"rank":1,"skierID":40213,"vertical":44120},{"rank":2,"skierID":11872,"vertical":43880}]}
< {"resortID":"7","serverTimestamp":"2026-10-16T21:50:40.052119Z","top":[{"rank":1,"skierID":40213,"vertical":44120},{"rank":2,"skierID":11872,"vertical":43880}]}
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zixin@Zixins-MacBook-Air ski-servlet %
```

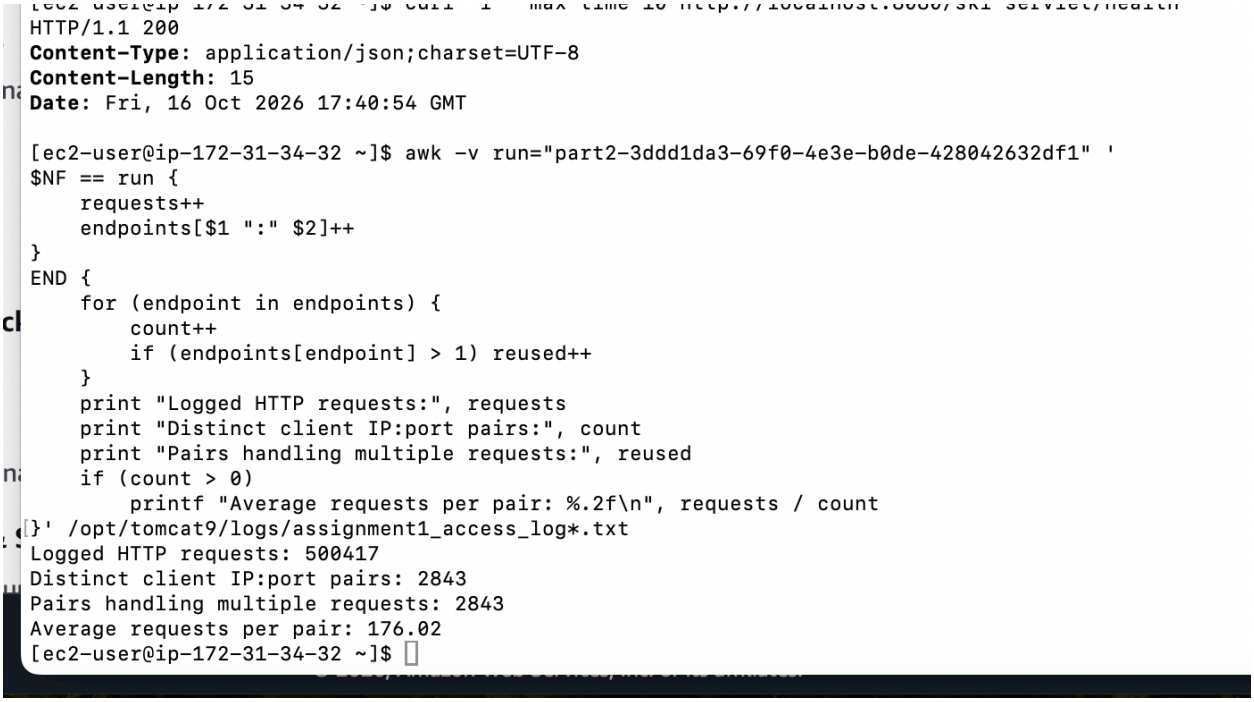
Connection Observation and Clock Limitations

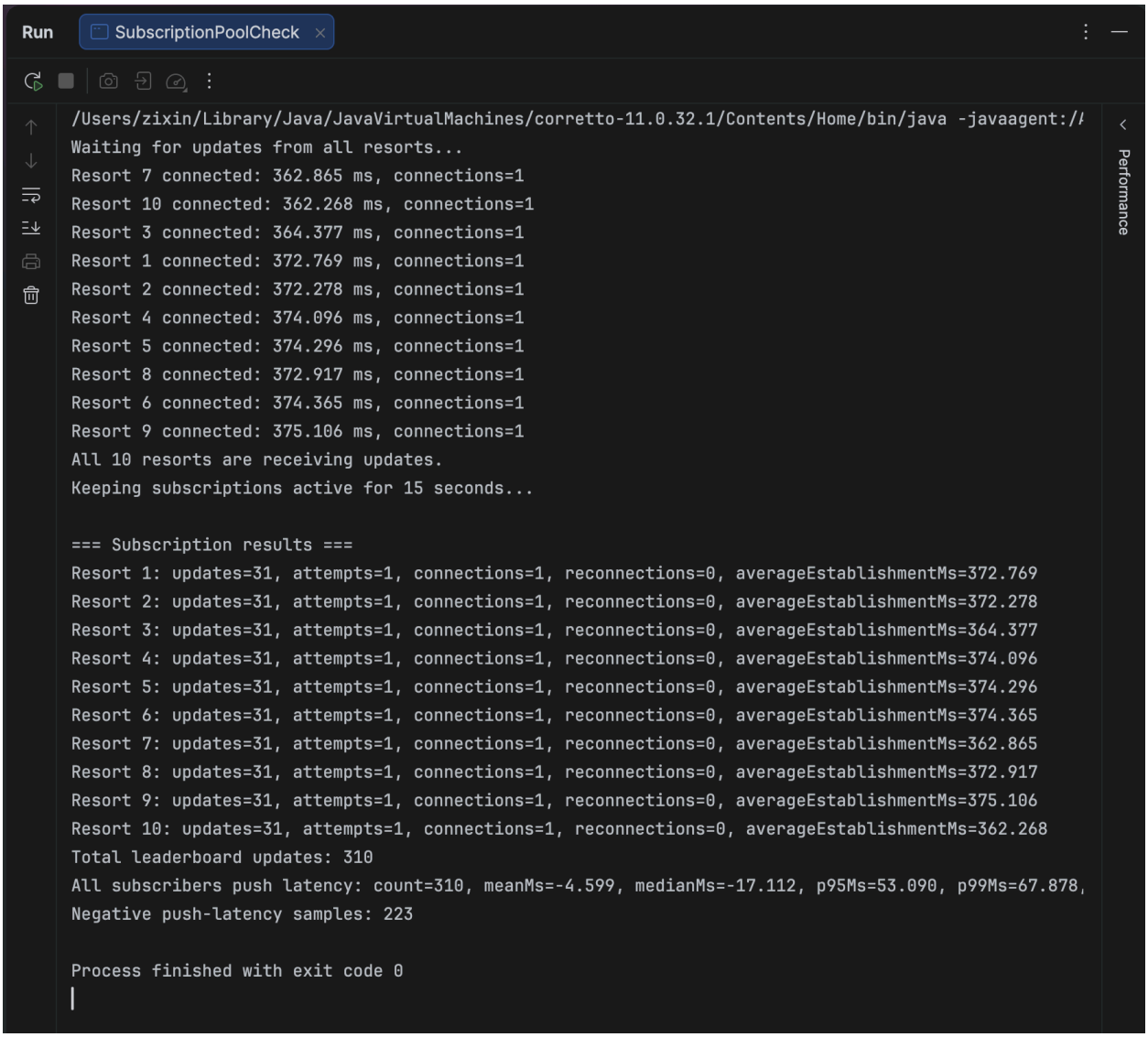
A separate 256-thread run attached an X-Run-ID header. Tomcat logged the client IP and port for matching requests.

The server recorded 500,417 requests across 2,843 distinct client IP:port pairs, averaging 176.02 requests per pair. Every observed pair handled multiple requests, supporting connection reuse. These pairs are not an exact TCP connection count, and HTTP retries are not reconnection counts.

Push latency uses the server timestamp and client receive time, so cross-machine clock error affects the result. Negative samples remained after Mac time synchronization. Raw values were retained without filtering or correction.

Push timestamp differences therefore cannot be interpreted as precise absolute one-way latency. Subscription-only clock checks are reported separately from HTTP load tests.





The HTTP connection observation and clock check were separate experiments from the 256-thread Run 2 performance test.